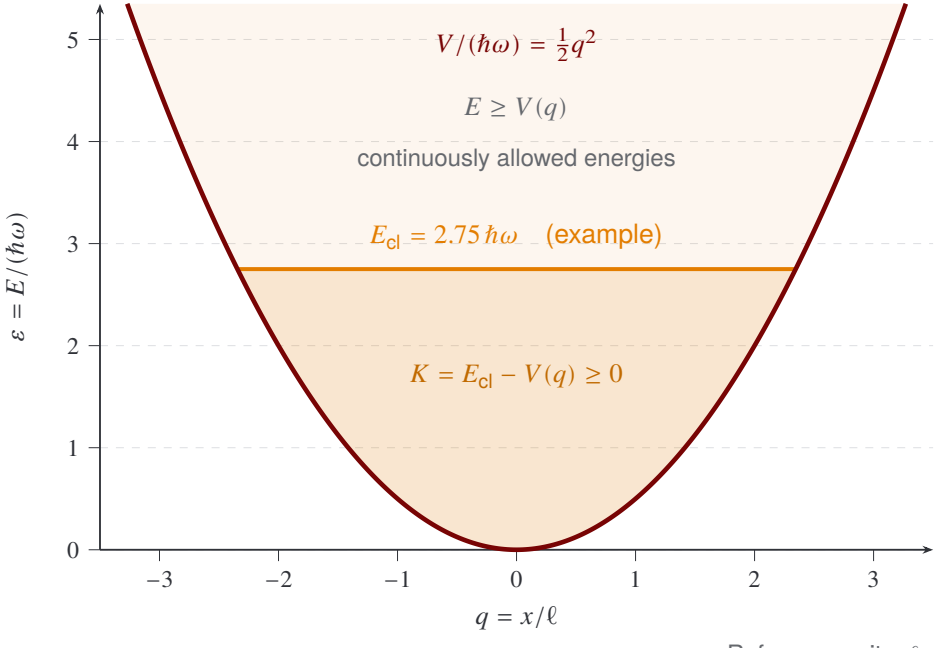
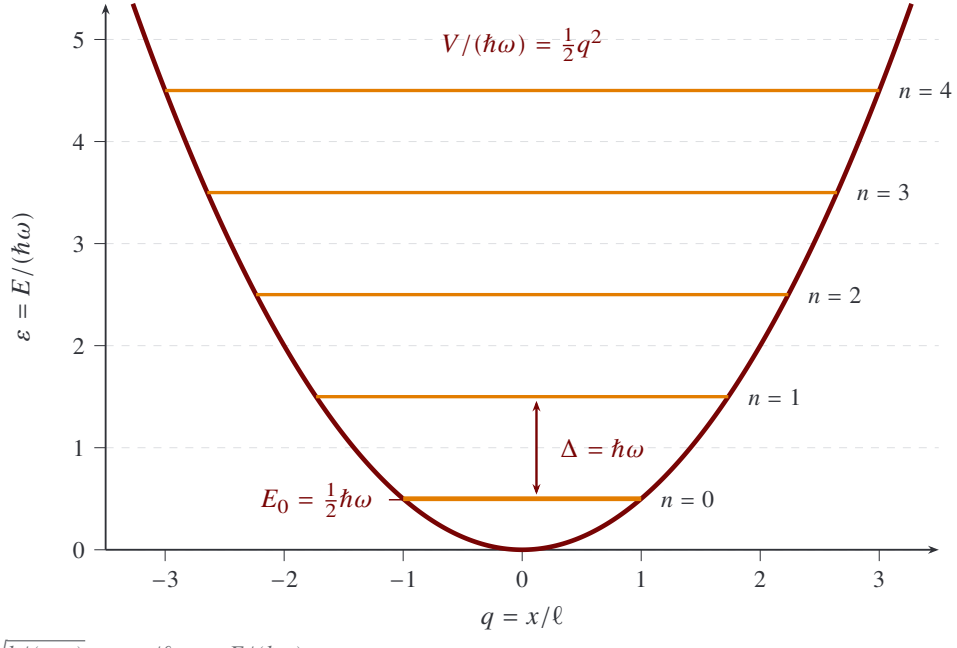
Classical harmonic oscillator

Continuous energy: any $E \geq 0$



Quantum harmonic oscillator

Discrete energy: $E_n = \hbar\omega(n + \frac{1}{2})$



Reference units: $\ell = \sqrt{\hbar/(m\omega)}$, $q = x/\ell$, $\varepsilon = E/(\hbar\omega)$.

$\hbar$ sets a shared plotting scale; it does not quantize the classical system.